

Road Network Planning Introduction

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1. Course Content
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- Understand the road network planning navigation feature

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- The core of the road network planning navigation feature is to use the Nav2 route server to help users create, edit, and manage the road network for robot navigation. A road network represents the valid paths a robot can follow in its environment and consists of nodes (waypoints) and edges (connections between waypoints). Unlike ordinary free-space planning, path-based navigation ensures that the robot follows specific, predefined paths.

Road network planning navigation is commonly used in the following scenarios:

- Industrial environments where specific routes must be followed
 - Use case: Factory AGV robots transport parts along a preset road network, with routes avoiding production lines, personnel walkways, and heavy equipment areas.
- Warehouse operations that require structured movement patterns
 - Use case: In a parts warehouse, the road network is divided into zones by material type. Robots pick and place parts along fixed routes to ensure first-in-first-out while avoiding shelf collisions.
 - Use case: Hotel service robots only travel along a specific route network when delivering goods to different rooms.
- Facilities with restricted areas or priority paths
- Large outdoor urban or natural environments
 - Use case: Route planning in navigation software is based on the city lane network, and autonomous sightseeing vehicles in scenic areas only travel along the park's fixed road network, avoiding ecological protection zones and dangerous road sections.

